

Business Process Compliance Tracking Using Key Performance Indicators

Azalia Shamsaei, Alireza Pourshahid, and Daniel Amyot

SITE, University of Ottawa, 800 King Edward,
Ottawa, ON, K1N 6N5, Canada
{asham092, apour024, damyot}@site.uottawa.ca

Abstract. Compliance of business processes with authoritative rules is significantly important to avoid financial penalties, efficiency problems, and reputation damages. However, finding the right measures to evaluate and track compliance is very challenging. We propose a novel method to model the context and measure compliance using the User Requirements Notation (URN). We mainly use Key Performance Indicator (KPI) extensions of URN to measure the level of compliance to rules. Such KPIs have been used in the past to measure the satisfaction level of goals and the performance of business processes. Yet, they have never been used for measuring compliance. Our method highlights the non-compliant policies and rules on a quadrant map based on their importance and compliance levels. Furthermore, we suggest a new method for importance calculation in this context. We use a human resource policy example to illustrate our method.

Keywords: Business Processes, Goal Modeling, Legal Compliance, Key Performance Indicator, Rules, User Requirements Notation.

1 Introduction

Compliance management is a critical activity in any organization. Every year, organizations invest time and money to ensure their processes comply with different regulations and policies. There are four different categories of *authoritative rules* [5] by which a business process can be governed. Each of these categories may be considered voluntarily or imposed to an organization. The first category is internal policies used in organizations to give better direction to business processes. Regulations and laws imposed by governments represent the second category. Violating rules from this category can cause important consequences, including financial penalties, loss of reputation, and lawsuits. The third category is composed of Service Level Agreements (SLA) used between organizations to ensure promised services are provided according to expectations. The last category of authoritative rules is that of standards.

With so many rules to follow, large organizations have a difficult time keeping track of compliance levels of different processes as well as of the organization as a whole. Some will only assess their compliance level based on the audits they pass or fail, an approach that is usually not very proactive and that may lead to some of the consequences mentioned earlier. One challenge is that it is next to impossible to

comply with all imposed rules given limited resources and rules that may conflict. Another is to keep the compliance level under continuous scrutiny to make sure the next important audit will pass.

In this paper, we propose a method enabling organizations to calculate their compliance level and discover business processes that violate authoritative rules. We use organization business goal, process and authoritative rule models as the input of our method. This method builds on the *User Requirements Notation* (URN), which is the first international standard to combine and integrate goal modeling (with URN's Goal-oriented Requirement Language — GRL) and scenario/process modeling (with URN's Use Case Map notation — UCM) [7,15]. URN models are graphical and can be created, managed and analyzed with jUCMNav [9], a free, Eclipse-based open source tool. The method outputs the compliance levels as well as target processes that require improvement. It takes advantage of an extension to URN used to model *Key Performance Indicators* (KPIs), which have been recently introduced for performance management [13], to calculate the compliance level of organizations. In addition, we define an algorithm that calculates the importance of low-level rules based on high-level goals and propagate the result to business processes and rules.

Related work in this area will be discussed in section 2, followed by an overview of our method in section 3. A case study from the human resource sector is elaborated in section 4 to illustrate our method and its benefits. Finally, the conclusion and future work will be discussed in section 5.

2 Related Work

Namiri and Stojanovic [12] have proposed a formal framework for managing business process compliance for enterprises. First, the accounts with a major impact on the enterprise's bottom-line are selected. Then, all relevant business processes are identified. Finally a set of controls is defined and tested regularly to protect the processes against identified risks. In addition, the framework suggests a set of properties to check the completeness of the compliance management system. These are essentially static constraints on the set of accounts, processes, controls, risks, and their relationships. This framework mainly focuses on compliance from a financial point of view for some of the business processes. We believe compliance monitoring should allow for all business processes and non-financial goals and constraints to be considered.

Kharbili and Stein [10] have proposed a policy-based semantic framework for enterprise compliance management. This framework defines a three-layer architecture and the transformations between these layers. The first layer uses semantic policies and rules to formalize regulations. In the second layer, these policies are transformed into business rules utilized in business process models. Finally, the business rules are transformed to operational rules used in automated business processes.

As suggested by Kharbili and Stein, a comprehensive compliance management framework should be integrated with business processes and support different levels of policy and rule definitions. Furthermore, according to zur Muehlen *et al.* [16], a combination of formal business process modeling and rule modeling language is required for proper documentation and modeling of compliance in an organization. Governatori *et al.* [4] go further by providing a heavily formalized, logic-based representation of rules and business processes where semantic annotations and model

checking are used to detect statically violations of obligations. Furthermore, Lu *et al.* [11] have also proposed a logic-based approach to evaluate compliance distance degrees (values between 0 and 1) between business processes and control objectives/rules. The end result of this method can be utilized by process designers to improve the compliance degree, but the complexity of the method may be an impediment for regular business users.

Although our proposed method does not provide the level of formalization and automated transformations of the above frameworks, the use of URN allows business users to model not only policies, rules, and processes but also other types of business and stakeholder goals. Hence, our approach supports business analysts in having a holistic view of the context. Furthermore, finding the source of problems and tracking compliance issues continuously becomes more systematic and better supported. Logic-based approaches have their own advantages and can perhaps be used to provide the theoretical background for the development of tools for compliance, but we believe that URN with KPIs adds a complementary and appropriate level of formalization in a business context, especially for end users. This combination is also more pragmatic for applications to actual running processes, not just to analysis of models.

Note that URN has already been used in the context of compliance. Ghanavati *et al.* [2,3] have proposed a URN-based requirements management framework for compliance management involving healthcare organization/legal goals (in GRL) and business processes (in UCM) linked to policies and legal documents. With this framework, they defined and analyzed different types of traceability and compliance links between organization models and regulation models. Although the modeling notation used in this framework is the same as in our proposed method, we not only establish traceability links between organization and regulation models but also *measure* the level of compliance of business processes against policies/regulations, with KPIs. Furthermore, we suggest a new method to calculate the relative importance of rules in this context.

3 Method Overview

The main objective of this method is to provide a modeling approach and guidelines enabling organizations to measure the current compliance level of their processes, track down and address compliance problems, and also evaluate the impact of changes on the overall compliance level of the organization. In order to meet this objective, the method requires the following inputs:

- The policy/regulation model, including sub-policies, rules and key performance indicators modeled with standard GRL extended with KPIs.
- The organization goal model (in GRL) and business process model (UCM).
- The importance level of high-level business goals.

The GRL policy/regulation model shows the high-level policies, their decomposition into sub-policies, and ultimately the operational rules stating how the processes can be compliant with the policies (see Fig. 6 and Fig. 7). Furthermore, we connect a set of KPIs that measure the level of compliance for the rules. Each rule can have one or more KPIs to measure its compliance level. The KPIs are defined by the analysts who

know both the business and the policies. Obviously, there may be situations where the required measures do not exist in current corporate data sources. Although this may seem to be a problem, this is usually a short-term one as such situation helps organizations detect inadequacies in their data sources and improve them to address their requirements for measuring compliance.

We also use business goals and business process models in our approach to make the analysis more accurate, based on facts, and provide better outcomes. The main usage of the business goal model is to specify the importance of the related business processes to each rule and policy and to illustrate the business context for the analysts. Measuring the importance of rules allows us to identify the most important problems that need to be fixed first. This is detailed further in section 4.2.

The expected outcomes of our method are:

- Compliance level measures for policies and rules.
- Location of policies/rules applied to the organization on a quadrant according to their compliance level and importance.
- Tracking compliance changes after business process or policy modifications.

As illustrated in Fig. 1, our method is iterative and consists of four main steps.

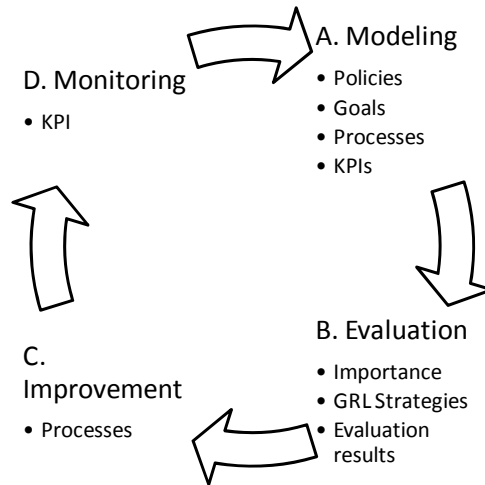


Fig. 1. Method steps

Step A. In the modeling step we:

1. Model the organization's business processes and business goals.
2. Associate processes to organization goals using URN links to define the importance level of processes as well as of the rules and policies applied to the processes.
3. Model policies, sub-policies and rules.
4. Associate rules to existing organization processes using URN links, to show where the rules are used in the organization.
5. Define key performance indicators for each rule.

Step B. In the evaluation step we:

1. Add GRL strategies (initial set of satisfaction values to some nodes in the goal model) and define values for the *target*, *threshold*, and *worst* attributes of KPIs. The values of KPIs' *evaluation* attributes come from external data sources. GRL's KPI construct maps the evaluation value to a satisfaction level (on a scale from -100 to 100, by linear interpolation considering the target, threshold, and worst values) that can then be propagated to other elements in the goal model according to the goal evaluation algorithms presented in [1].
2. Define the importance level of high-level business goals and calculate the importance of process and rules accordingly. The importance of the rules should be calculated considering both the importance of related business goals and the policies/regulations that the rule is part of. At this point, we have only considered business goal importance related to a rule and the algorithm result is added to the model manually (see section 4.2), but in the future we plan to provide an automatic algorithm propagation to make this effort easier.
3. Evaluate the compliance (i.e., the satisfaction level of policies)
 - I. Illustrate policies on a quadrant diagram according to their compliance level (satisfaction level) and importance value.
 - II. Highlight the critical rules with low satisfaction levels and track down the associated processes using the URN traceability links defined in step A.

Step C. In the improvement step, we change processes to address the problems highlighted in the quadrant diagram. We also evaluate whether our modifications to fix one problem affect negatively the compliance against other rules/policies. This is done again with GRL strategies, where we can simulate many what-if situations.

Step D. In the monitoring step, we monitor the compliance KPIs to observe whether the expected changes actually happen. If the results are not the ones expected, we take corrective actions, otherwise we can move on to improve other necessary processes.

4 Human Resource Case Study

In our case study, we focus on a human resource policy described in [6]. This particular policy is divided into three groups totalizing 36 sub-policies. The main policy we focus on is the hiring policy. In the rest of this section, we go through the steps of the method presented in section 3 (except the monitoring step due to space limitation) and explain them in more detail in the context of the case study.

4.1 Step A: Modeling

In this first step, we concentrate on providing the inputs required for performing the analysis. These inputs include the business process model (see Fig. 2, Fig. 3, and Fig. 4) in UCM, the organization goal model (see Fig. 5), and the policy model (see Fig. 6 and Fig. 7) in GRL.

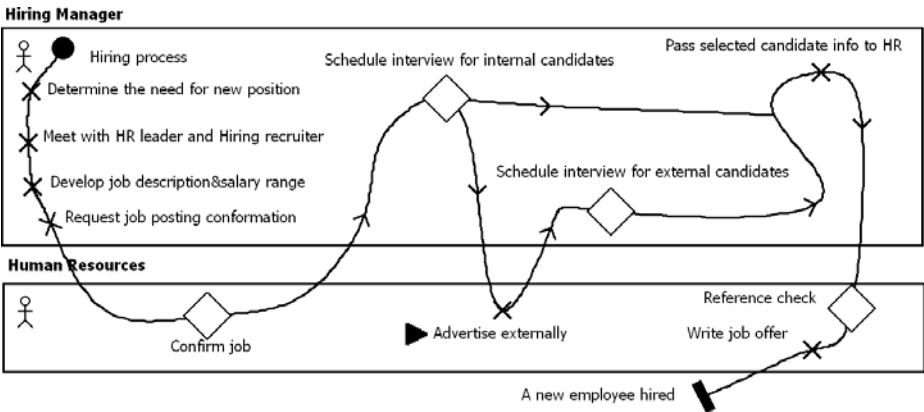


Fig. 2. Hiring process (UCM)

UCM models (see Fig. 2) consist of causal sequences of responsibilities that can be allocated to components. A scenario path is triggered by a start point (●, e.g., Hiring process) and results in an end point (■, e.g., A new employee hired). Responsibilities (×, e.g. Determine the need for new position) can be defined along the way to represent actions or tasks performed during the process. UCM supports both concurrent (⊢) and guarded alternative (≡) paths. Furthermore, stubs (◇, e.g., Confirm job) contain sub-maps used to organize complex processes in hierarchies of sub-processes (e.g. Fig. 3 and Fig. 4). Finally, actors (□, e.g., Hiring Manager) can be used to separate different parts of a process based on organizational units or roles.

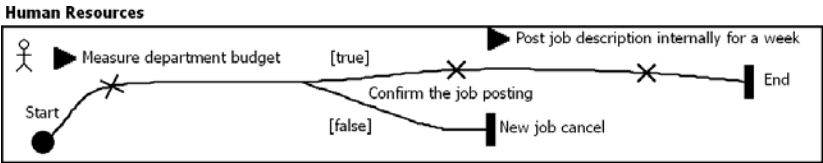


Fig. 3. Confirm job sub-process (UCM)

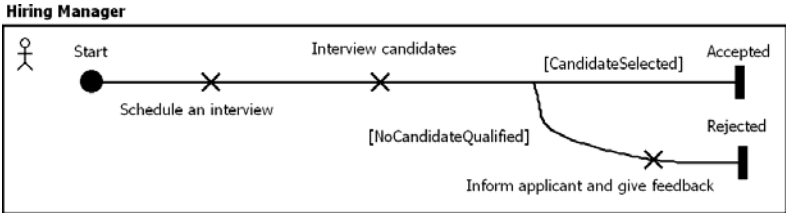


Fig. 4. Schedule interview for internal candidate sub-process (UCM)

In the hiring process case study, a Hiring Manager determines the need for a new job position and discusses the job description and salary with Human Resources. It is Human Resources' responsibility to confirm a new job while considering the

department's budget. Upon successful confirmation, Human Resources post the job internally for a week and the Hiring Manager schedules an interview with successful internal candidates. If no internal candidate is qualified for a job then interviews will be scheduled for external candidates until a successful candidate is found.

In the next step, we model the organization's high-level goals. GRL models (Fig. 5 to Fig. 7) consist of *intentional elements* such as softgoals ($\square$, e.g., Successful business), goals ($\square$, e.g., HR Policies) and KPIs ($\diamond$). Intentional elements can be assigned to stakeholders through actors ($\odot$). In addition, intentional elements can be connected to each other using different types of links, including AND/OR decomposition (not used here) and weighted contribution links ($\rightarrow$). The latter indicate the positive or negative impact of intentional elements on each other (with a quantitative scale from 100 to -100), which is used for evaluations and analysis. The model elements explained above are the subset or URN used in this paper. Both UCM and GRL have more model elements used in other contexts (see [7] for a complete list).

Fig. 5 illustrates the parts of the organization goals we are interested in. This model is useful for analysts to know the context and better prioritize the processes and rules according to the organization goals. The high-level goal Successful Business is part of our method for calculating the importance level.

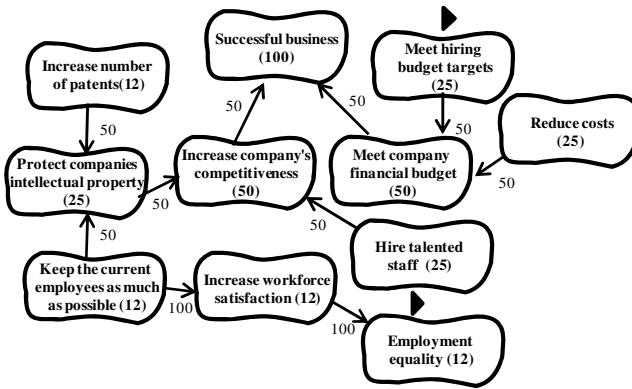


Fig. 5. Organization's high-level business goal model

Policies and regulations can be modeled in a similar way. For example, we have modeled 5 sub-policies for HR Policy with GRL (Fig. 6). We assume that all sub-policies have an equal impact on HR Policy, and hence the contribution links have a weight of 20. We continue modeling sub-policies until we reach the rule level. Due to space limitation, we only indicate sub-policies, rules and key performance indicators for the Hiring sub-process. Fig. 6 shows HR Policies (at the top) and sub-policies for the Hiring Policy (at the bottom). Fig. 7 models rules and key performance indicators for the job posting policy. We have assigned an identifier to each rule (R1 to R4) for reference in the rest of the paper.

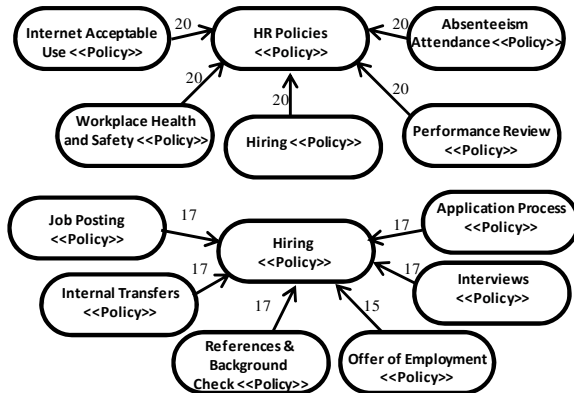


Fig. 6. Sub-policies for Human Resources (top) and Hiring (bottom)

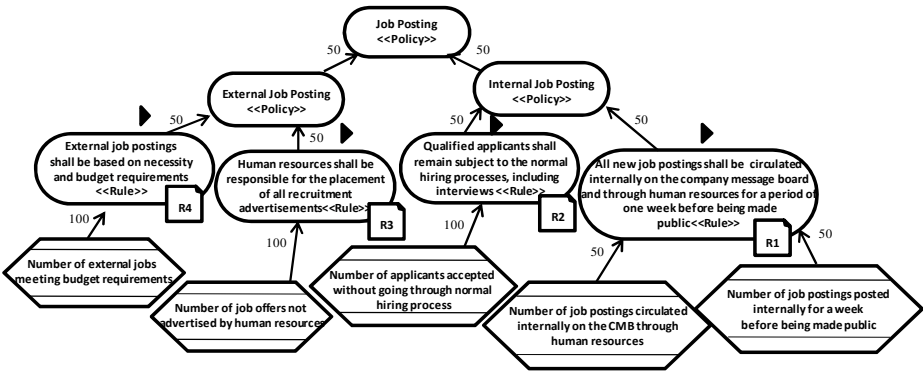


Fig. 7. Rules and key performance indicators for Job Posting

Table 1. Mapping rules to business processes

Business Process	Rule
Determine whether the department can afford hiring employees	R4
Post job description internally for a week	R1
Advertise externally	R3
Schedule an interview for internal candidates	R2

We then map rules to business processes with *URN links* (whose presence is indicated by a black triangle). Table 1 shows the traceability links between business processes and rules in our case study. These links allow us to highlight the processes that need to be modified in the improvement step, if any. Furthermore, they allow us to identify the importance of rules based on their associated business processes.

4.2 Step B: Evaluation

In this step, we define the GRL *strategies* used to evaluate the satisfaction level of policies. Each GRL strategy initializes the KPI value sets [13]. Table 2 shows the values for a sample strategy we have used in this scenario. The evaluation values can be entered manually like other value sets (which is useful for the analysis of what-if situations) or automatically obtained from a data source such as a Business Intelligence system. These strategies are used to evaluate the satisfaction level of the other intentional elements (i.e., rules and policies) using the standard quantitative evaluation method supported by GRL and jUCMNav [1].

Table 2. List of key performance indicators for rules

Rule	Key Performance Indicator	Target	Threshold	Worst	Evaluation
R1	Number of job postings circulated internally on the CMB through human resources	10	10	0	10
	Number of job postings posted internally for a week before being made public	10	10	0	5
R2	Number of applicants accepted without going through normal hiring process	0	0	10	6
R3	Number of job offers not advertised by human resources	0	2	10	1
R4	Number of external jobs meeting budget requirements	10	10	0	8

In addition to the satisfaction value, the *importance* value (to the containing actor/stakeholder) is also calculated using a new algorithm for GRL intentional elements in this step. As illustrated in Fig. 5, *Successful business* is the highest goal in the organization model has an importance value of 100. The algorithm will calculate the importance for other intentional goal using their contribution links' values. Importance values are propagated down from high-level goals to business processes and rules according to the following formula:

$$\text{Importance Value} = (\text{Parent Importance Value}) * (\text{Proportion of Contribution Level})$$

For instance, the importance value of R4 is propagated down all the way from the highest level goal of the business (see Fig. 5). *Success business's* importance value is 100 and it is connected to *Meet company financial budget* through a contribution link with a contribution level of 50. Therefore, the importance value of *company financial budget* becomes 50. Subsequently, the importance value of *Meet hiring budget target* (the goal linked to the process to which R4 is applied) becomes 25. This value is propagated to business processes and rules through URN links.

Using these two values, the rules will be laid out on a quadrant as shown in Fig. 8. A similar quadrant diagram was used before for performance monitoring [14]. However, using the quadrant in the context of policy monitoring is another application explored here. This quadrant allows us to *identify what is the most important rule that the organization is not compliant with*. This will allow the people in charge to more easily prioritize the improvement candidates for the next step.

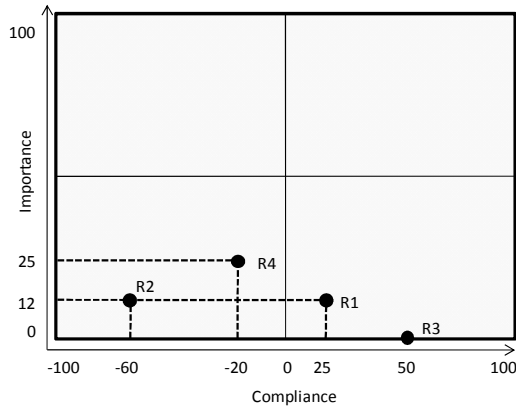


Fig. 8. Importance and compliance of rules on a quadrant diagram

4.3 Step C: Improvement

As shown in Fig. 8, the compliance levels of R2, R4, R1 and R3 have the highest to lowest levels of non-compliance, in that order. In terms of importance levels, R4 is the most important rule while R3 is the least important. Rule R2 has the highest level of non-compliance and its importance is in the same quadrant area as R4. Even if there is a slight different between their importance levels, since R2 has the highest non-compliance value, it takes priority over R4. Therefore, we should try to improve the process associated to R2 in order to get the highest impact on our global compliance level.

As the KPIs related to R2 indicate, out of ten internal applicants in a year, six were employed without being interviewed first. We suggest here two modifications to this business process model to better enforce the rules and increase the compliance:

- Human Resources shall be involved in scheduling and performing interviews.
- Interviewers shall fill out the job candidate evaluation form while performing interviews.

By adding these two steps to the business process, the number of candidates hired without interviews will be reduced. When comparing the new process with the previous one, this time not only the hiring manager will be part of interview process but also will human resource personnel. Furthermore, each employee should have a job candidate evaluation form that will be filled out by interviewer team during interview.

Before making any concrete change to the processes, we also estimate its cost to the organization. A human resource manager working in Canada earns an average salary of \$83,859 [8]. We have assumed there are five one-hour interviews per job. Hence, the total cost of a human resource manager participating in 10 interviews would be about \$2966 (assuming 49 weeks * 37.5 hours/week of work, plus 30% of overhead). Although the suggested modification will improve the compliance of the current business process, it also clearly increases its cost. However, in this case, the cost is not really that high and it is worthwhile to increase the chances of hiring an

external talented candidate (who could be productive well over the additional cost during his/her career) by adding the extra suggested steps to the business process.

5 Conclusions and Future Work

Today's organizations cannot comply fully with all the various rules imposed on them, and hence they should comply with the rules that affect their business the most. In this paper, we defined a method that enables compliance tracking for business processes using key performance indicators. Organization goals and processes are modeled using the same lightweight language (URN) as policies and rules. This method helps finding how well authoritative rules are satisfied based on the importance and compliance values for the rules. The importance values are propagated down from high-level business goals to business processes and rules. Also, the non-compliant business processes are highlighted for further analysis and actions. In addition, our method is not limited to one aspect of the organization (e.g., financial). Although this is not apparent from our small example, the goals of all the organization's stakeholders can be considered during the analysis using GRL actors. These capabilities are quite powerful and allow one to have a holistic view of the organization.

Although our method is based on URN capabilities, other process (e.g., BPMN or YAWL) and goal modeling languages (e.g., *i** or TROPOS) could also be used to address the same issue. However, at this point we are not aware of any other language that provides the same level of support for modeling and analyzing goals and processes in an integrated manner, especially when KPIs need to be considered.

This method also suffers from several limitations that require further investigation. As we mentioned earlier, the importance value of each rule is based on the business goal importance set by the organization's managers. We may not have a business goal for all rules. We should be able to set the importance from the policy/regulation side as well. This is very important in the context of regulations. Furthermore, this example only has one business process per rule, which makes improvements very localized. In more complex situations, multiple rules may apply to a particular process, and hence a change to improve the satisfaction of one rule may have side effects on the satisfaction of the other rules. This will require i) making assumptions about the impact on the KPIs (to simulate whether the change can lead to a globally better set of satisfactions, and this can be done with jUCMNav), and ii) monitoring the situation to make sure the desired effect is really happening (step D of the method).

Acknowledgments. This research was supported by NSERC's Business Intelligence Network.

References

1. Amyot, D., Ghanavati, S., Horkoff, J., Mussbacher, G., Peyton, L., Yu, E.: Evaluating Goal Models within the Goal-oriented Requirement Language. *International Journal of Intelligent Systems* 25(8), 841–877 (2010)
2. Ghanavati, S., Amyot, D., Peyton, L.: Towards a Framework for Tracking Legal Compliance in Healthcare. In: Krogstie, J., Opdahl, A.L., Sindre, G. (eds.) *CAiSE 2007 and WES 2007*. LNCS, vol. 4495, pp. 218–232. Springer, Heidelberg (2007)

3. Ghanavati, S., Siena, A., Perini, A., Amyot, D., Peyton, L., Susi, A.: A Legal Perspective on Business: Modeling the Impact of Law. In: 4th Intl. MCeTech Conf. on eTechnologies. LNBIP, vol. 26, pp. 267–278. Springer, Heidelberg (2009)
4. Governatori, G., Hoffmann, J., Sadiq, S., Weber, I.: Detecting Regulatory Compliance for Business Process Models through Semantic Annotations. In: Business Process Management Workshops (BPM 2008). LNCS, vol. 17, pp. 5–17. Springer, Heidelberg (2009)
5. Hamou-Lhadj, A.K., Hamou-Lhadj, A.W.: Towards a compliance support framework for global software companies. In: 11th IASTED Int. Conf. on Software Engineering and Applications, pp. 31–36. ACTA Press, Anaheim (2007)
6. HRdownloads Website (2010), <http://www.hrdownloads.com/>
7. International Telecommunication Union: Recommendation Z.151 (11/08), User Requirements Notation (URN) – Language definition, <http://www.itu.int/rec/T-REC-Z.151/en>
8. Job Canada, <http://www.jobcanada.org/salary.html> (last accessed April 15, 2010)
9. jUCMNav, Version 4.2.1, University of Ottawa (2010), <http://softwareengineering.ca/jucmnav>
10. Kharbili, M.E., Stein, S.: Policy-Based Semantic Compliance Checking for Business Process Management. In: MobIS Workshops 2008. CEUR Workshop Proceedings, vol. 420, pp. 178–192 (2008)
11. Lu, R., Sadiq, S., Governatori, G.: Measurement of Compliance Distance in Business Processes. *Information Systems Management* 25(4), 344–355 (2008)
12. Namiri, K., Stojanovic, N.: Towards A Formal Framework for Business Process Compliance. In: Multikonferenz Wirtschaftsinformatik (MKWI 2008). GITO-Verlag, Berlin (2008)
13. Pourshahid, A., Amyot, D., Chen, P., Weiss, M., Forster, A.J.: Business Process Monitoring and Alignment: An Approach Based on the User Requirements Notation and Business Intelligence Tools. In: 10th Workshop on Requirements Eng. (WER 2007), pp. 80–91 (2007)
14. Pourshahid, A., Chen, P., Amyot, D., Forster, A.J., Ghanavati, S., Peyton, L., Weiss, M.: Business Process Management with the User Requirements Notation. *Electronic Commerce Research* 9(4), 269–316 (2009)
15. Weiss, M., Amyot, D.: Business process modeling with URN. *International Journal of E-Business Research* 1(3), 63–90 (2005)
16. zur Muehlen, M., Indulska, M., Kamp, G.: Business Process and Business Rule Modeling Languages for Compliance Management: A Representational Analysis. In: ER (Tutorials, Posters, Panels & Industrial Contributions) 2007. ACM Int. Conf. Proceeding Series, vol. 334, pp. 127–132. Australian CS, Darlinghurst (2007)